

Muhammad Ramis

Security Engineer

OSCP-Certified Security Engineer | Application Security | 8 Years Engineering Experience

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PROFILE

OSCP-certified security engineer with 8 years of engineering experience, including dedicated security ownership across high-volume web applications, APIs, and Linux systems. Operated in environments handling up to 6.5M API requests per day, 2.8M monthly active users, and multi-terabyte customer data stores. Combines offensive testing, secure code review, AI-assisted triage, and root-cause analysis to build durable, automated security solutions directly into engineering workflows.

CORE SKILLS

Assess and Strengthen: Penetration testing, vulnerability assessment, exploit validation, OWASP Top 10, API security

Attack and Defend: Threat modelling, authentication and authorization testing, access control, secure code review

AI and Automation: Python, Bash, LLM triage, embeddings, AI agents, SAST, DAST, continuous testing

Privacy and Compliance: GDPR safeguards, privacy-by-design, data-flow mapping, least privilege, encryption, audit trails

Engineering: JavaScript, Node.js, React, PHP, SQL, Linux, Docker, GitHub Actions, CI/CD

AI SECURITY AUTOMATION PROJECT

Creator and Lead Developer | BugTraceAI | 2026

Stack: Python, FastAPI, Go, Playwright, React, PostgreSQL, LLM agents

- Designed a self-hosted AI security scanning platform with a 6-phase pipeline for discovery, analysis, consolidation, exploitation, validation, and reporting.
- Built specialist agent workflows across 14 vulnerability classes, combining Go fuzzers, Playwright browser checks, and LLM-guided payload mutation.
- Added consensus voting, circuit breakers, and authenticated scanning, publishing a demo with 145 findings and 43 validated issues.

EXPERIENCE

Security Engineer | Sorvox Groups | 2022–2024

Stack: Python, Bash, Linux, Burp Suite, REST APIs, Docker, CI/CD, SAST/DAST

- Baseline-tested 5 inherited product stacks after acquisition, covering 42 services, 180 endpoints, 2.8M monthly users, and 9TB of data.
- Built LLM triage with embeddings over past findings, CWE, and OWASP notes, classifying 8,400 weekly scanner results and cutting duplicate review by 61%.
- Traced authentication and authorisation paths from routes to database permissions, proving 17 exploitable access-control flaws and driving fixes into regression tests.
- Mapped GDPR Article 30/32 controls to logging, retention, and encryption checks, closing 24 privacy-control gaps before integration.
- Automated endpoint enumeration, dependency checks, and DAST smoke tests in CI/CD, reducing manual assessment setup from 2 days to 3 hours.

Full Stack Developer | Apricart | 2020–2022

Stack: React, Node.js, PHP, Laravel, SQL, REST APIs, Docker, GitHub Actions

- Built and secured e-commerce APIs serving 6.5M daily requests, 450k active customers, 1.2M catalogue items, and 2.3TB of MySQL data.
- Reworked customer-data and admin API flows after modelling trust boundaries, closing 31 IDOR, SQL injection, XSS, and CSRF risks pre-release.
- Blocked 127 vulnerable package releases with SAST, dependency scanning, peer-review gates, and release checks across 14 repositories.
- Mapped 74 PII fields across checkout, CRM, and admin exports, adding audit logs, retention checks, and access controls for GDPR DSAR readiness.

Software Engineer | Creatie Drop | 2018–2019

Stack: PHP, JavaScript, MySQL, Linux, REST APIs, Git, Web Security

- Maintained production web platforms handling 900k monthly sessions, 70k registered users, 28 integrations, and 380GB of transactional data.
- Mapped API failures across controllers, SQL queries, and access checks to eliminate unsafe trust assumptions in payment and account flows.
- Implemented server-side validation, authentication checks, and defensive error handling, cutting high-priority production bugs by 37%.

Software Developer | Chatcloud | 2016–2018

Stack: JavaScript, PHP, MySQL, Linux, APIs, Git, Secure Coding

- Built messaging features for 55k registered users, 1.1M monthly messages, 120k daily webhook events, and 180GB of MySQL data.
- Used root-cause debugging across APIs, authentication, and data workflows to remove recurring security failure modes from release paths.
- Added structured logging, safer error handling, and access reviews, reducing escalated incidents by 29%.

CERTIFICATIONS

OSCP - Offensive Security Certified Professional | 2025

Practical penetration testing certification covering exploitation, privilege escalation, reporting, and remediation.

EDUCATION

MSc Cyber Security and Artificial Intelligence | The University of Sheffield, UK | Sep 2024–Nov 2025

Distinction; dissertation on post-quantum blind signatures with Python, Rust, Node.js, gRPC, and REST tooling.

BSc Computer Science | PAF KIET, Karachi, Pakistan | Aug 2014–Dec 2018

First Class; final project: React Native carpooling platform with web admin, GPS, chat, and Laravel backend.

SELECTED ACHIEVEMENTS

- **OffSec The Gauntlet Echo Response S1:** Placed 4th in a forensics CTF by analysing evidence and correlating artifacts under time pressure.
- **ShefESH Capture The Flag:** Won 1st place by exploiting web vulnerabilities, documenting impact, and explaining remediation paths.
- **ACM-ICPC Regional Finals:** Placed top 10 and fastest solver by decomposing unfamiliar algorithmic problems against 40+ regional teams.
- **Technical Leadership:** Co-led programming competitions for 100+ participants; mentored peers on debugging, algorithms, and secure logic.